

Chandan Kumar

AI Engineer • LLMs, Real-Time Voice AI, Computer Vision

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SUMMARY

Founding AI Engineer at OSVI.ai, building the core agent engine of a real-time voice AI platform. Previously shipped computer-vision identity verification at NeuroFin.ai and ML systems at Turocrates.ai (IIT Bombay). Merged open-source work in Microsoft DeepSpeed, CMU's ESPnet and Alibaba's SenseVoice; IEEE-published author.

TECHNICAL SKILLS

LLMs & GenAI	RAG (LangChain, LlamaIndex, ChromaDB), AI agents, tool calling, MCP, LoRA/PEFT fine-tuning, LLM evals
Speech & Voice AI	LiveKit voice agents, speech-to-speech (Gemini Live, OpenAI Realtime), Whisper, ESPnet, VAD
Deep Learning & CV	PyTorch, Hugging Face Transformers, TensorFlow, CNNs, YOLO, PaddleOCR, ArcFace, spaCy
Inference & MLOps	llama.cpp (GGUF quantization), DeepSpeed, CUDA, FastAPI, Docker, Kubernetes, AWS, GCP Vertex AI
Languages	Python, TypeScript, Go, C/C++ , SQL

EXPERIENCE

OSVI.ai Aug 2026 – Present

Founding AI Engineer

- Building the core agent engine behind every call on OSVI's voice AI platform: ASR, LLM reasoning, TTS and telephony.
- Built voice-to-voice calling on Gemini Live and OpenAI Realtime, cutting first-response latency, with fallback to the STT-LLM-TTS pipeline so calls survive model outages.

NeuroFin.ai Sep 2025 – Jul 2026

AI Engineer

Bengaluru, Onsite

- Built a fintech KYC system that classifies documents, extracts fields with OCR and matches faces to ID photos.
- Hardened the pipeline against spoofed scans and low-light selfies, reducing false rejects within product latency SLAs.

Turocrates.ai (IIT Bombay) Dec 2025 – Apr 2026

ML Engineer

Remote

- Owned the model lifecycle: data preparation, training, evaluation and low-latency inference APIs for product teams.
- Built evaluation harnesses and regression baselines so model upgrades could be compared, debugged and rolled out safely.

IIT Ropar, Annam.ai Jun 2025 – Jul 2025

AI Research Intern

Ropar, Punjab

- Built a ResNet-50 plant-disease classifier (93% accuracy, 87K+ images) served via FastAPI with batched inference, plus a recommender with NLP sentiment analysis over agricultural news.

ISRO, Liquid Propulsion Systems Centre Jun 2024 – Jul 2024

Machine Learning Intern

Kerala, Onsite

- Built a YOLOv5 and PaddleOCR pipeline that digitised 500+ legacy engineering drawings; denoising, adaptive thresholding and deskewing improved robustness by 15%, and my CRNN vs Tesseract vs PaddleOCR benchmark set the team's OCR stack.

OPEN SOURCE (15 MERGED PULL REQUESTS, GITHUB.COM/MODELPATH-DEV)

- Microsoft DeepSpeed** (43k stars): the RL rollout engine now rejects shared prefill under continuous batching instead of silently ignoring it. **Microsoft APM**: fixed version-tag resolution for marketplace installs.
- ESPnet, Carnegie Mellon University** (10k stars, 2 PRs): rebuilt convolutional-subsampling pad masks from each utterance's own length, so batched ASR decoding now matches single-utterance decoding.
- Alibaba SenseVoice** (9k stars): fixed device auto-detection so the ASR container starts on CPU-only machines. **LLamaSharp** (3 PRs): added CUDA 13 builds and runtime probing.

PROJECTS

ShieldPrompt: PII Masking for LLMs | *Python, FastAPI, Presidio, spaCy NER, MCP* GitHub • PyPI

- Masks PII with reversible vault tokens before prompts reach LLMs; shipped as a library, FastAPI middleware, CLI and MCP server for Claude Code and Cursor. Drew seed-funding interest from Apertu Capital.

BitWiser: 1-bit LLM Compression on Apple Silicon | *llama.cpp, Metal, FastAPI, React* GitHub

- Importance-matrix-calibrated IQ1_S quantization shrinks Llama-3.1-8B from 16.1 GB to 2.19 GB (7.4x), running at ~51 tokens/s in 3.1 GB of RAM, with a live side-by-side benchmark playground.

Mistral-7B Domain Q&A with RAG | *Hugging Face, LoRA/PEFT, 4-bit quantization, LangChain, ChromaDB* GitHub

- Fine-tuned Mistral-7B on 10K+ Q&A pairs with LoRA, cutting perplexity by 35% and the 4-bit model from 14 GB to 4 GB at under 2% accuracy loss; the RAG pipeline reached 91% retrieval accuracy.

EDUCATION

Vellore Institute of Technology (VIT), Vellore *B.Tech, Computer Science and Engineering* 2022 – 2026

ACHIEVEMENTS

- Published Paper**, IEEE PICC 2025, "Automated Ranking of Video Frames Based on Clarity" Oct 2025
- 7th Rank**, CodeChef-VIT Hackathon, top 2% of 400+ teams (2023) • **Outreach Head**, IEEE SPS, VIT (2024)